

University of East London

School of Architecture, Computing and Engineering

Lab Report

CN6005 Artificial Intelligence

Module Tutors: Dr. Maimoona Sharif

Module Leader: Dr. Nadeem Qazi

Academic Year: 2026 – 2027

Week 2: Implementing Explainable Linear Regression using SHAP

Student ID: 2923566

Student Name: Mazen Kozbar

- # Introduction:

This project predicts the iris flower target from four measurements: sepal length, sepal width, petal length and petal width. A linear regression model is used because it is simple and easy to interpret, which suits the second aim of the project: understanding why the model makes the predictions it does. To do this, the model is evaluated with mean squared error and R^2, and then explained with SHAP, which shows how much each measurement pushes a prediction up or down.

- # Method:

The iris dataset was loaded and split into the four measurements as inputs and the target as the output. The data was then divided into training and testing sets, with 20% held back for testing, so the model could be judged on flowers it had never seen. A fixed random seed of 42 was used so the split, and therefore the results, can be reproduced. A linear regression model was trained on the training set and used to predict the test set. Mean squared error was chosen to measure how far the predictions were from the true values on average, and R^2 to show how much of the variation in the target the model explained. For the explanation, a SHAP explainer was built around the model's predictions using the training data as the background, and applied to the test set. A SHAP summary plot was then produced to compare the features, along with a dependence plot for each of the four measurements to see how each one affects the prediction individually.

- # Explanation:

The model performed well, with a mean squared error of 0.04 and an R^2 of 0.95, so it explained about 95% of the variation in the target on the test set. The SHAP summary plot shows where this comes from. Petal length and petal width have by far the largest impact: high values of both push the prediction up and low values push it down. Sepal length has a smaller effect in the opposite direction, with higher values slightly lowering the prediction, and sepal width has the least influence, since its points sit close to zero. This means the model relies mainly on the petal measurements, and the sepal measurements add little by comparison. This only reflects the 30 flowers in the test set, so it should be read as a pattern in this dataset rather than a general rule.